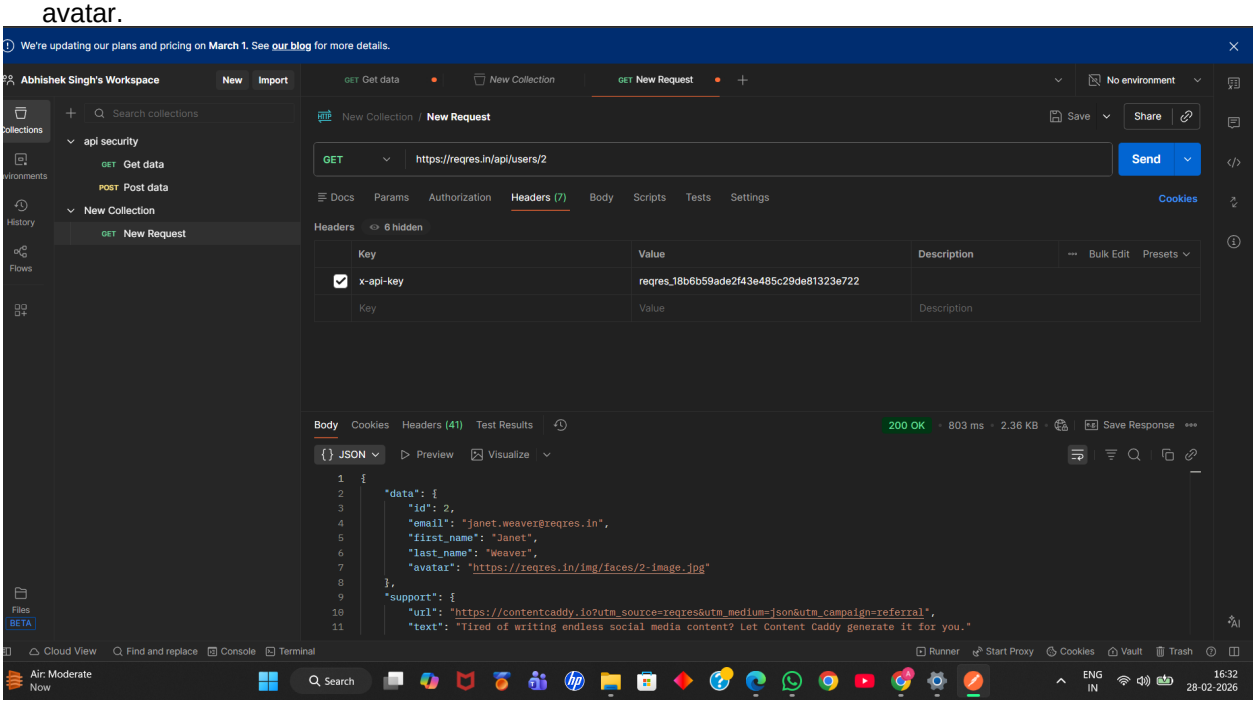
API Security Testing Report

1. Introduction

This report documents API security testing performed using Postman on the ReqRes API platform. The objective was to validate authentication, request headers, API key usage, and response handling.

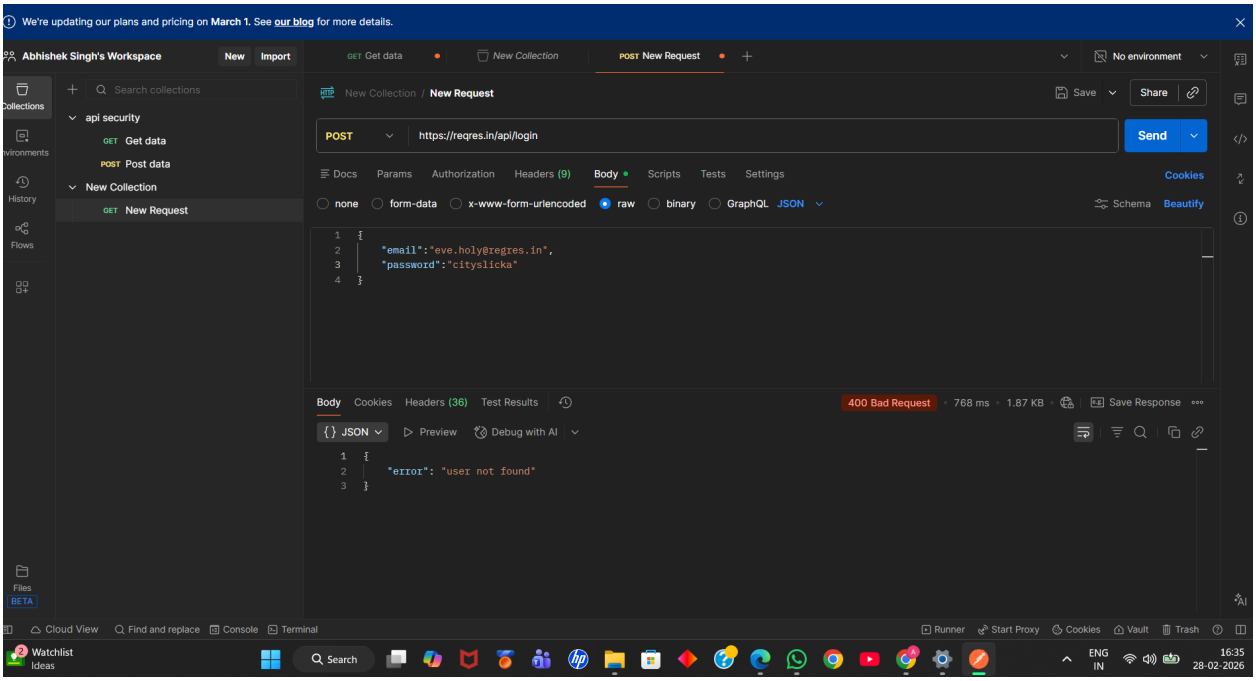
2. GET Request with API Key

Endpoint: https://reqres.in/api/users/2
Method: GET
Header Used: x-api-key
Result: 200 OK response received with user data including id, email, first name, last name, and avatar.



3. POST Login Attempt (Invalid User)

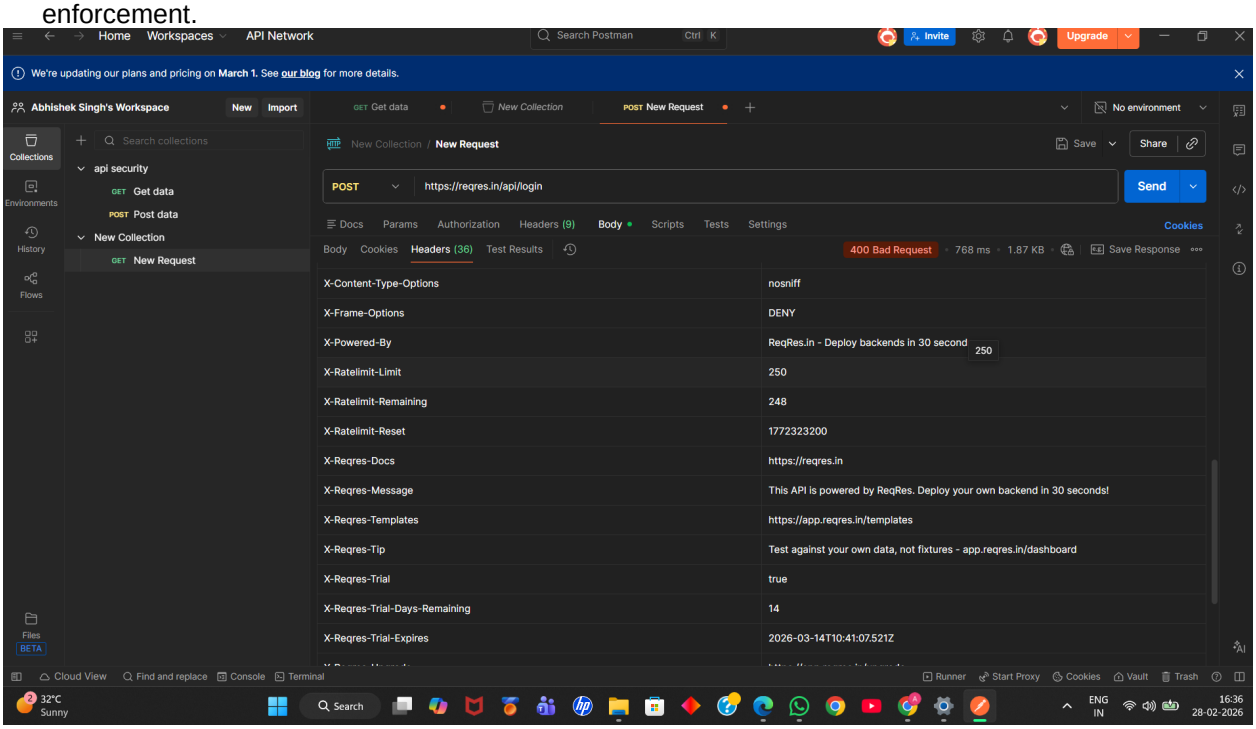
Endpoint: https://reqres.in/api/login
Method: POST
Body (JSON): email and password provided.
Result: 400 Bad Request with error message 'user not found'.



4. Response Headers Analysis

Observed security-related headers:
- X-Content-Type-Options: nosniff
- X-Frame-Options: DENY
- X-RateLimit-Limit: 250
- X-RateLimit-Remaining: 248

These headers indicate protection against MIME sniffing, clickjacking, and rate limiting enforcement.



5. Conclusion

The API security testing confirms correct implementation of authentication validation, header-based API key handling, and security response headers. Error handling for invalid login attempts is functioning correctly with appropriate HTTP status codes.